
Is There a Universal ‘Sounds Like AI’ Direction? Probing and Steering AI Style in Transformer Residual Streams

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Abstract

Detection of AI-generated has become increasingly important as large language models are deployed at scale. Prior work has shown that high-level concepts such as truthfulness, refusal, and sentiment are encoded as linear directions in the residual streams of transformer models. We investigate whether “sounding like AI” — the composite stylistic quality that makes LLM outputs identifiable — is similarly represented, and whether steering along this direction can evade real-world AI detectors. Using contrastive activation addition (CAA) on Qwen2.5-3B, we extract an “AI style direction” from the HC3 dataset. Replicating prior work, we confirm that this direction achieves 97.5% classification accuracy on ChatGPT-sourced text, though 93% of the signal overlaps with text length. After controlling for length, an 85.5% accurate residual style direction remains. Steering along this direction reduces Pangram’s AI probability from 0.79 to 0.02 on ChatGPT-sourced text, demonstrating near-complete evasion of a state-of-the-art detector. We further extend the dataset to include Claude and Gemini responses, finding that these models produce substantially longer outputs (97.5% and 96.1% length overlap respectively), which largely dominates the extracted direction. To address this, we propose a sentence-boundary truncation method that reduces length-direction cosine similarity from 0.944 to 0.002, revealing a genuine cross-source style signal at 95.8% accuracy. However, steering on this length-controlled combined dataset produces inconsistent evasion results, suggesting that a universal model-agnostic AI style direction may not be straightforwardly steerable. Our results highlight both the promise and the limitations of residual stream manipulation as an approach to AI detection evasion.

1 Introduction

2 Introduction

The proliferation of large language models has intensified an ongoing arms race between AI text generation and AI text detection. Tools such as GPTZero, ZeroGPT, and Pangram attempt to identify machine-generated text by exploiting statistical and stylistic signatures in LLM outputs, while a parallel line of work explores evasion strategies. Existing evasion approaches largely operate at the surface level: recursive paraphrasing Sadasivan et al. [2023], post-hoc humanization, or prompt engineering Mitchell et al. [2023]. These methods treat the model as a black box, modifying outputs after the fact without any understanding of *why* the text sounds AI-generated in the first place. We ask a more fundamental question: is “sounding like AI” encoded as an internal representation within the model itself, and if so, can we manipulate that representation directly to evade detection?

A growing body of work in mechanistic interpretability has established that transformer language models linearly encode high-level concepts as directions in their residual streams Marks and Tegmark

[2024], Arditi et al. [2024], Konen et al. [2024], and that these directions can be extracted and causally steered via Contrastive Activation Addition Rinsky et al. [2024]. Whether “sounding like AI” — the composite stylistic quality that makes LLM outputs identifiable — constitutes such a direction has not been directly investigated. Concurrently, Kuznetsov et al. (2025) use Sparse Autoencoders on residual streams to identify features contributing to AI text detection, finding that style and verbosity features are the most discriminative. Our work is complementary but distinct: we use the simpler difference-in-means approach and target *evasion* rather than detection interpretability.

We extract an “AI style direction” from Qwen2.5-3B using the HC3 dataset of paired human and ChatGPT responses, replicating prior work showing 97.5% classification accuracy with 93% overlap with text length. After controlling for length, a genuine residual style signal persists at 85.5% accuracy. Critically, steering along this direction reduces Pangram’s AI probability from 0.79 to 0.02, demonstrating near-complete evasion of a state-of-the-art detector through internal representation manipulation. We then extend to Claude and Gemini, finding that these models’ substantially greater verbosity dominates the extracted direction even more severely. To address this, we introduce a sentence-boundary truncation method that eliminates the length confound by construction, revealing a genuine cross-source style signal at 95.8% accuracy — but finding that steering on this combined dataset produces inconsistent evasion results, suggesting that a universal model-agnostic AI style direction may not be straightforwardly steerable.

Our main contributions are:

- We demonstrate near-complete evasion of Pangram (0.79 $\rightarrow$ 0.02 AI probability) via activation steering on ChatGPT-sourced text, establishing residual stream manipulation as a viable attack vector against AI detectors.
- We systematically characterize the length confound across three AI sources (ChatGPT, Claude, Gemini), showing verbosity accounts for 93–97.5% of the extracted direction and is substantially more severe for Claude and Gemini.
- We introduce a sentence-boundary truncation method that reduces length-direction cosine similarity from 0.944 to 0.002, isolating a genuine cross-source style signal at 95.8% accuracy.
- We provide evidence that while a cross-source AI style direction exists as a classifier, it does not steer cleanly, suggesting AI style may be partially model-specific rather than a universal property of LLM-generated text.

3 Related Work

Linear representations in residual streams. A growing body of work in mechanistic interpretability has shown that transformer language models encode high-level concepts as linear directions in their residual streams. Marks and Tegmark Marks and Tegmark [2024] provide foundational evidence, demonstrating that truth and falsehood are linearly represented and extractable via difference-in-means probing — the core technique we employ — with causal verification via surgical intervention in the forward pass. Arditi et al. Arditi et al. [2024] extend this to behavioral properties, showing that refusal in safety-tuned models is mediated by a single direction whose ablation disables refusal and whose addition induces it on harmless inputs. Rinsky et al. Rinsky et al. [2024] introduce Contrastive Activation Addition (CAA), the methodology directly underlying our work, which extracts steering vectors by averaging residual stream differences between contrastive pairs and applies them at inference time with a tunable coefficient. Konen et al. Konen et al. [2024] further demonstrate that style properties — sentiment, emotion, writing style — can be extracted and steered via activation engineering without any training. Our work asks whether “sounding like AI” is similarly encoded, extending this line of research to a stylistic property with direct practical implications for detection evasion.

AI text detection and evasion. Mitchell et al. Mitchell et al. [2023] introduce DetectGPT, a zero-shot detector exploiting the observation that LLM-generated text occupies negative curvature regions of a model’s log probability function. Sadasivan et al. Sadasivan et al. [2023] stress-test the robustness of a broad range of detectors — including watermarking, neural network-based, and zero-shot approaches — showing empirically and theoretically that existing methods are unreliable under paraphrasing attacks. Our approach is fundamentally different from such surface-level evasion:

rather than modifying outputs after generation, we intervene directly on the model’s internal style representation during the forward pass.

Residual stream features for AI text detection. Most closely related to our work, Kuznetsov et al. [2025] apply Sparse Autoencoders (SAEs) to Gemma-2-2b’s residual stream to identify features contributing to AI text detection, finding that style and verbosity features are the most discriminative. They also find, consistent with our results, that text length is a major confounding factor. Our work differs in three key ways: we use the simpler difference-in-means approach rather than SAEs; we target evasion rather than detection interpretability; and we evaluate against real-world detectors including Pangram.

4 Background

5 Methods

6 Experiments and Results

7 Discussion Conclusion

Summary of our findings:

Implications

Conclusion

8 References

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